A Comparative Study of the Wake Effects Between the 2025 and the 2026 Formula 1 Models using Computational Fluid Dynamics (CFD)

How does the new 2026 Formula 1 model compare to the 2025 model in terms of its Wake Effects?

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November 2025

Abstract

This paper investigates how the aerodynamic wake produced by the 2026 Formula 1 model differs from that of the 2025 model under the new FIA regulations published on the 24th of June 2024 and the edits made on 31st July 2025. Using Computational Fluid Dynamics (CFD) simulations conducted in SimScale, the study compares drag coefficients, wake velocity profiles, and turbulence intensities between both models. The goal is to determine whether the new 2026 regulations successfully reduce wake turbulence and allow for closer racing. Results are expected to show a shorter, less energetic wake in the 2026 model, validating the effort done by the Federation Internationale de l'Automobile (FIA) to improve aerodynamic efficiency and racing dynamics.

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Introduction

The aim of this paper is to determine whether the 2026 new regulations will benefit closer racing or not by comparing both the 2025 and the 2026 regulations. Closer racing occurs when a trailing car can remain in the wake of a leading car without significant losses in downforce (Villanueva et al., 2020). To achieve this, the leading car must generate a wake that has a high momentum and has fewer large vortices, also, it has to be replaced by fresh ambient air (Mobility Engineering Technology, 2019). Lastly, the trailing car must be designed to efficiently operate within this disturbed inflow. (Pérez Nebot et al., 2025; Deepak, 2024)

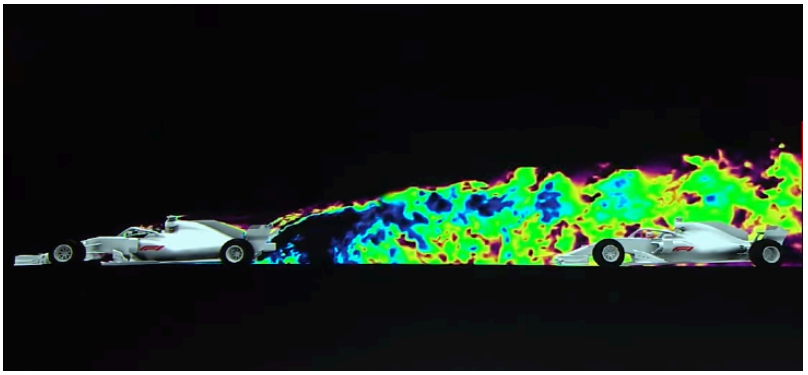


Figure 1) Wake Effect shown on a following car (AWS)

In Formula 1, eleven teams compete against each other each year to win the world championship. The FIA is the regulator of the competition, and for a set amount of years, they create the FIA regulations. These regulations dictate very demanding conditions that the teams must meet in order to compete. If they fail to comply with the FIA's regulations, they will not be able to race, or if found out after the race, they will be disqualified. Therefore, teams must design their cars in order to maximize performance while meeting with all the standards set by the FIA. (Suttil, 2024)

In the year 2022, a new generation of Formula 1 cars was introduced due to a new regulatory period; these new cars were aerodynamic masterpieces. (Fédération Internationale de l'Automobile, 2021) The 2025 car, which was of the same nature as the 2022 car (as there hadn't been any regulatory changes), was designed to conduct the airflow under the car, generating a vacuum that multiplied downforce. (SkySports, 2022) Therefore, cars could hold higher cornering speeds, due to the greater centrifugal force generated by the downforce. However, this design had a major flaw, the car produced an enormous quantity of wake effects, reducing the drag of the following car and not allowing overtakes. (Tshikalange, 2023, p. 7)

In 2024, a new regulation proposal was sent to the teams, "The 2026 F1 Season Technical Regulations". This proposal expects to allow closer racing and maxime show, thanks to various changes such as more agile cars (30 kg lighter), a redesigned power unit, active aerodynamics, an additional battery deployment system, improved safety and a record number of six power unit manufacturers.

In this study, the active aerodynamics and the new shape of the cars will be the main focus, especially as they replace the current Drag Reduction System (DRS), which provides an additional 10-12km/h of top speed. (F1 Chronicle, 2024) The new active aerodynamics and the shape of the cars are expected to not only be more sustainable, in line with F1's Net-Zero 2030 objective (Formula 1, 2025), but also be faster, with Toto Wolff (Mercedes AMG Team Principal) and George Russell (Mercedes AMG Driver) claiming that the 2026 cars could achieve 400km/h in long straights after trying the new 2026 car in the simulator. (SoyMotor.com, 2025)

The comparison is between the current car generation (2022-2025) and the one being introduced next year (2026). The reason for that is that the question to be answered only involves if the FIA has assessed their main problem, and doesn't focus on the wake effects over different periods, but rather on the wake evolution between 2025 and 2026.

Aerodynamics is the study of forces and the resulting motion of objects that travel through air. (NASA, 2023) To calculate values related to aerodynamics, we use Computational Fluid Dynamics (CFD). "CFD is a numerical simulation tool that enables the study of fluid flow and heat transfer phenomena in various engineering applications. It is widely used in industry and academia as a cost-effective and efficient approach to analyze complex fluid dynamics problems. CFD uses mathematical models and algorithms to solve the governing equations of fluid motion, including the conservation of mass, momentum and energy. In F1, CFD has become an essential part of the design process, allowing teams to develop and test new ideas quickly and efficiently in a virtual environment before implementing them on the physical car." (Méndez, 2023, p. 2)

What causes the reduction of the wake effect?

The wake effect refers to the region of disturbed airflow that forms downstream of a moving vehicle, characterised by reduced velocity, low static pressure, and increased turbulence intensity. In Formula 1, this wake significantly affects the performance of a following car by reducing the available downforce, thereby limiting grip and overtaking opportunities. Understanding what causes a reduction in the wake effect therefore requires an analysis of how aerodynamic design influences drag, flow separation, turbulence generation and pressure recovery behind the car (Katz, 2016).

From a physical perspective, the wake is a direct consequence of momentum loss in the airflow as it passes around the vehicle. According to fluid dynamics theory, aerodynamic drag is associated with a deficit in momentum within the wake, meaning that a larger drag force corresponds to a stronger and more persistent wake (NASA Glenn Research Center, n.d.). Drag force can be expressed as

$$F_D = \frac{1}{2} \cdot \rho \cdot v^2 \cdot C_d \cdot A \quad (1)$$

Where:

F_d — the drag force [N]

ρ — the density of the fluid [$kg \cdot m^{-3}$]

v — the speed of the object relative to the fluid [$m \cdot s^{-1}$]

C_d — the drag coefficient [*unitless*]

A — the cross sectional area of the object [m^2]

Consequently, reducing either C_d or the effective cross-section area, or both, reduces the magnitude of the wake (NASA Glenn Research Center, n.d.). This relationship provides the physical justification for using drag-related quantities as indicators of wake strength (See equation 2 in the Methodology)

Wake formation is also strongly influenced by flow separation, which occurs when the boundary layer loses sufficient kinetic energy to remain attached to the surface. Separation causes large-scale vortices and a sharp pressure drop at the rear of the car, significantly increasing pressure drag. Pressure drag arises from the imbalance

between high pressure at the front of the car and low behind it, which can be described quantitatively through Bernoulli's principle:

$$p + \frac{1}{2}\rho \cdot v^2 = \text{constant} \quad (2)$$

And can be seen in Figure 2)



Figure 2) Visual representation of a wing's flow separation (Wikipedia)

Although real F1 flows are viscous and turbulent, this relationship highlights that regions of high velocity correspond to lower static pressure. When flow separates, low-pressure zones persist behind the car, strengthening the wake (Katz, 2016). Aerodynamic designs that delay separation allow the airflow to maintain higher velocity and pressure coherence downstream, thereby reducing wake intensity.

Ground-effect aerodynamics play a crucial role in reducing this wake disturbance. Downforce generated by the underfloor relies on accelerating airflow beneath the car and expanding it smoothly through the diffuser. The diffuser's role is to recover pressure by allowing the

high-speed underbody flow to decelerate gradually. If the diffuser angle is too steep, flow separation occurs, increasing turbulence and wake strength. Efficient diffuser design improves pressure recovery, which can be linked to a smaller pressure gradient ∇p behind the car. CFD pressure contours show that better diffuser performance results in higher static pressure immediately downstream, indicating a weaker wake (Katz, 2016).

It is important to note that steady-state Reynolds-Averaged Navier-Stokes (RANS) simulations rely on turbulence modelling assumptions and cannot capture all unsteady phenomena. However, for comparative aerodynamic studies, RANS methods reliably identify relative changes in drag, pressure recovery and wake structure (Versteeg & Malalasekera, 2007). Therefore, the CFD results obtained in this investigation should not be interpreted as absolute values, but rather as comparative between both models.

In conclusion, the reduction of the wake effect is caused by a combination of decreased pressure drag, improved flow attachment, lower turbulence generation, enhanced pressure recovery and efficient ground-effect aerodynamics. These mechanisms reduce the momentum deficit and energy dissipation in the airflow, producing a narrower, less turbulent wake that is less detrimental to following vehicles. Therefore, the ultimate goal of this investigation is to take into account some of these factors and find a “Wake Rating” to compare both car models.

Note that due to limited time available for this investigation, the CFD simulations were restricted to a reduced number of configurations and conducted using steady-state RANS models rather than more computationally intensive methods. Consequently, certain factors

that influence the size and structure of the aerodynamic wake could not be taken into account in this investigation, and will be proposed as a possible extension.

Methodology

This paper will compare the two models according to a "Wake Rating" determined by the following formula:

$$Wake\ Rating\ = C_d \cdot A \cdot p_{max.} \quad (3)$$

Where:

C_d — drag coefficient [unitless]

A — cross-section area of the car [m^2]

$p_{max.}$ — pressure felt by the rear-wing [Pa]

This Wake Rating considers three main factors that result from the CFD analysis that will be carried out.

The first factor is the drag coefficient (C_d), a dimensionless number that tells how much aerodynamic resistance an object experiences as it moves through a fluid, in this case, air. A lower C_d indicates less drag, and therefore a faster and more efficient car. A higher C_d indicates more drag, and therefore a slower and less efficient car.

The second factor is the cross-section area of the car (A), in other words, the frontal area that faces the airflow when the car is moving forwards. For a Formula 1 car, the typical value oscillates around 1.5 and 2.0 m^2. As the cars do not have a plain surface, rather a complex one, calculating A without using CFD is not accurate enough, as the shapes of the car are to be approximated using simpler shapes such as rectangles, trapezoids and triangles.

The last factor is the maximum pressure felt by the rear-wing. This measure indicates how big or small the pressure is at the rearmost part of the car. The least pressure, the least drag will the trailing car get, and the bigger the pressure, the more drag will the trailing car experience.

The two cars will be compared using the Wake Rating, the car with a higher rating will have a more energetic, turbulent airflow, therefore disturbing the other car's expected laminar airflow, and reducing its grip on the asphalt.

Experiment Conditions

The Computational Fluid Dynamics (CFD) analysis will be carried out in SimScale. The model used in the investigations belong to the SimScale repertory, concretely "F1 2026 Regulation Car" by reesewu2005. The computational domain will be a rectangular wind tunnel with the dimensions seen in Table 1), such dimensions are enough to avoid interference according to a study by Jichao Li et al. The computational domain is summarized in Table 1). (Li et al., 2025)

Dimension	Value
Upstream distance	4 car lengths
Downstream distance	6 car lengths
Side clearance	1.5 car widths
Height clearance	6 car heights

Table 1) CFD Domain Dimensions

The mesh type is triangular, the mesh includes the external flow volume and the car itself. In the case of the 2026 car, the meshed model has 25.8M cells and 7.8M nodes. The fineness of the model is set at 5 (which is the average fineness allowed by SimScale, that allows 1-10).

Physics Models

In the simulation, the air density was 1.1965 [kg/m^3] and the viscosity was 0.000015295 [m^2/s]. The turbulence model used was Newtonian, as it is the most accurate representation of the situation simulated in this study. The flow is steady-state, as F1 aerodynamics are often analysed with steady-state RANS for comparative studies.

The 2026 model is (5.252 x 1.148 x 1.88) metres in size, smaller than the 2025 model, as a consequence, it will make the cross-section area of the 2026 car smaller than the 2025 model.

Boundary Conditions

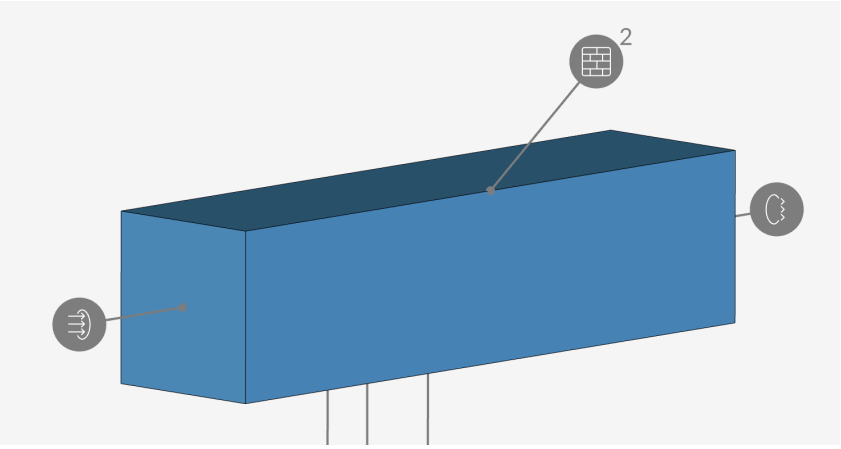


Figure 3) Boundary conditions of the simulations (SimScale)

For the simulation, the velocity inlet was 100m/s, equivalent to 360km/h. This velocity was chosen as it corresponds to top-speed conditions reached by modern F1 cars on long straights (in circuits such as Monza, Baku, Spa-Francorchamps…) At these speeds, aerodynamic forces dominate vehicle behaviour, making it an appropriate regime for analysing drag and wake effects. This happens because drag scales with the square of velocity ($F_d \propto v^2$), so differences between aerodynamic configurations (i.e. the differences between the 2025 and the 2026 car) become most pronounced. Moreover, at 360km/h, wake structures and pressure deficits are fully developed, allowing clear differentiation between both geometries, more stable force convergence and as a result, a more meaningful wake analysis.

The pressure outlet was set to 0 Pa gauge pressure, allowing all the airflow coming from the inlet to escape freely, without affecting the measurements on the car itself.

The side and top walls of the external box are set as no-slip walls. The bottom wall is set to travel at 100 m/s, to mimic a moving car. The tyres are also set to rotate at the same speed.

Results

To find the value of the wake rating for both cars, we need the values for $C_d \cdot A \cdot p_{max.}$, and as SimScale only outputs the C_d value and does not output the other quantities directly, those will need to be derived to calculate the Wake Rating for the 2026 car.

Note that to calculate the Wake Rating for the 2025 car, data from previous studies will be used due to time and computer power constraints.

The cross-section area for the 2026 model will be determined from the CFD force outputs provided by SimScale. The solver reports the aerodynamic drag force acting on the body in the flow direction as a force component F_D (in newtons), which was extracted from the Forces and Moments results for each of the two simulation runs. Since SimScale does not directly output cross-section areas, it will be derived from Equation (1) rearranged for A

$$F_D = \frac{1}{2} \cdot \rho \cdot v^2 \cdot C_d \cdot A$$

$$A = \frac{2 \cdot F_D}{\rho \cdot v^2 \cdot C_d} \quad (4)$$

Note that v is the relative velocity of the car with respect to air (considering the frame of reference shown in Figure 3), therefore:

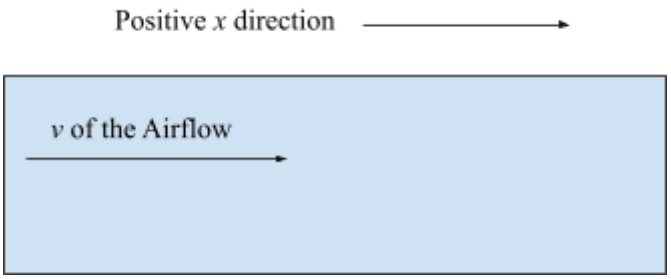


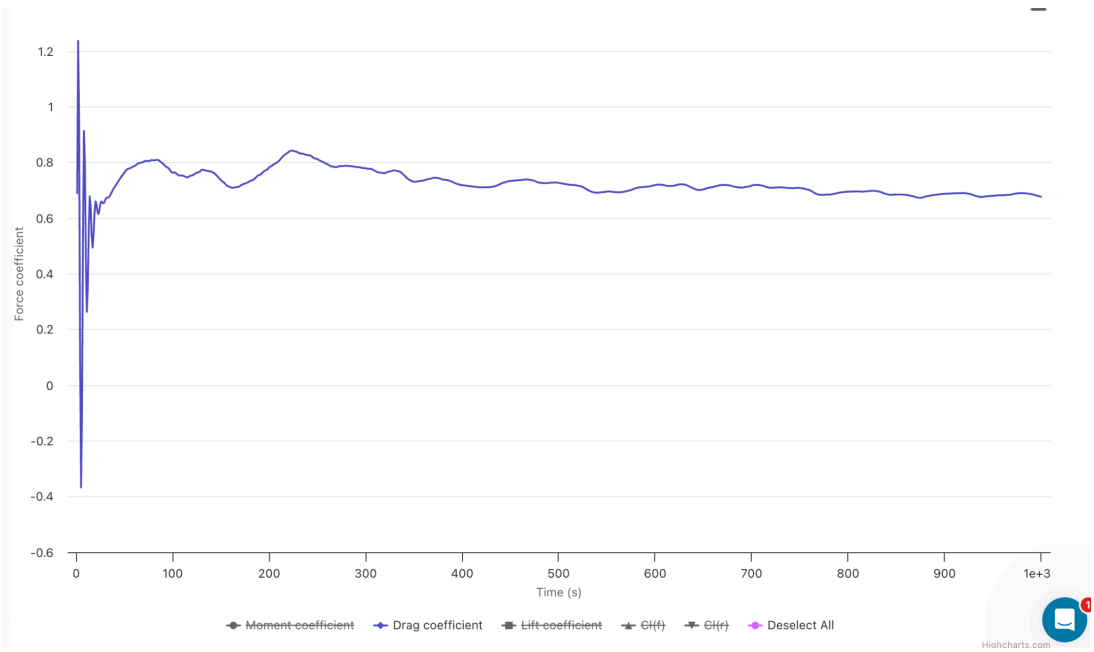
Figure 4) Relative velocity diagram (self-drawn)

$$v_{relative} = v_{air} - v_{car}$$

$$v_{relative} = 100 - 0\, m \cdot s^{-1}$$

$$v_{relative} = 100\, m \cdot s^{-1}$$

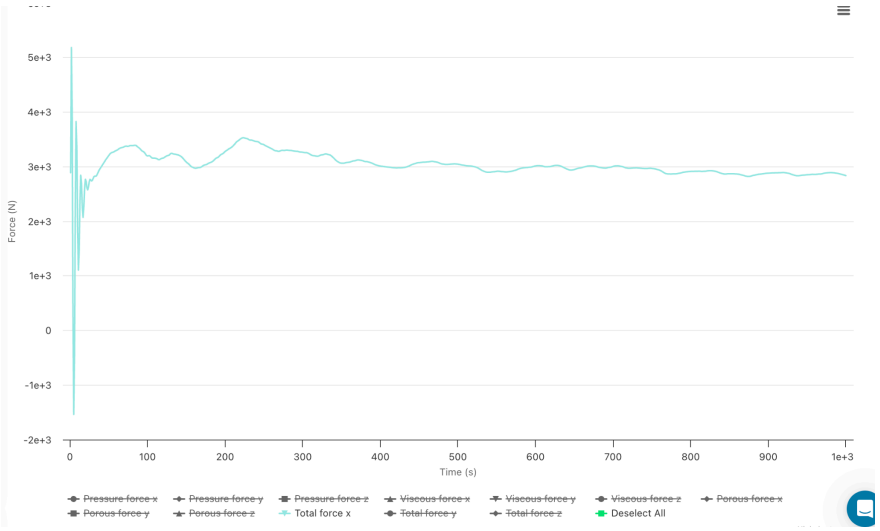
The drag coefficient (C_d) for the 2026 model is shown in the following graph, which shows the C_d for every iteration.



Graph 1) Drag Coefficient for the 2026 model

Therefore, for the 2026 Model, the resultant drag coefficient, taken from the last iteration, is 0.67765.

The Drag Force in the x direction felt by the 2026 model is shown in the following graph:



Graph 2) Total Drag Force for the 2026 model

Therefore, for the 2026 model, the total drag force felt by the car in the x direction is $2,836.81\ N$.

Substituting these values into equation (4) yields the value for A in the 2026 car.

$$A = \frac{2 \cdot F_D}{\rho \cdot v^2 \cdot C_d}$$

$$A = \frac{2 \cdot 2{,}836.81}{1.1965 \cdot 100^2 \cdot 0.67765}$$

$$A = 0.699\ m^2$$

Data extracted from the CFD Simulations:

Data	2026 Model
F_D	2,836.81 Newtons
C_d	0.67765 [unitless]
$v_{relative}$	$100\ m \cdot s^{-1}$
ρ	1.1965 $[kg/m^3]$

Table 2) Data extracted from the CFD simulation

Finally, the last term of the Wake Rating is the maximum pressure felt by the rear-wing.

According to SimScale, this is the centre of the rear-wing, as can be seen in Figure 5)

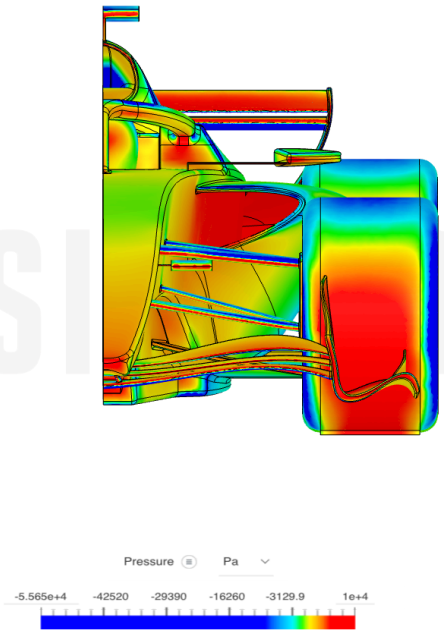


Figure 5) Cross-section view of the 2026 Model

According to the program, the pressure at the centre of the rear-wing is 4589.86 Pascals

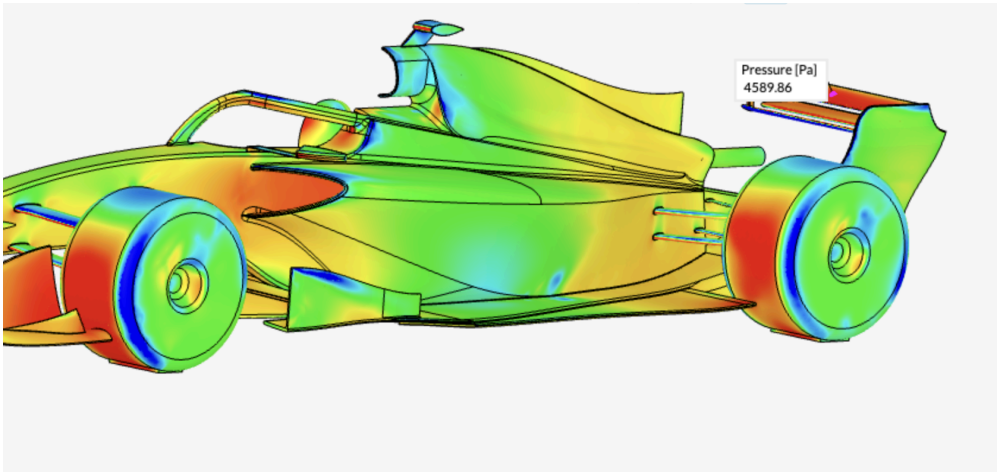


Figure 6) Pressure at the centre of the rear-wing.

With all the information above, the Wake Rating for the 2026 Model can be calculated following equation (3):

$$Wake\ Rating\ = C_d \cdot A \cdot p_{max.}$$

$$Wake\ Rating\ = 0.67765 \cdot 0.699 \cdot 4,589.86$$

$$Wake\ Rating\ = 2,174.11$$

Having conducted the CFD Simulation, a comparison between the simulation and a study made on a 2022-regulation car (meaning that it is legal from 2022 to 2025) will be done.

2022 Regulated car data

Data	2022 Regulated Model
C_d	1
A	0.744
p_{max}	4589.86*

Table 3) Data for the 2022-regulated cars

The cross-sectional area of the 2022 car was calculated by scaling A_{2026} up by the size difference between the two cars. Assuming the same height, the width from the model is 1.88 m wide and the 2022 rules allow 2.000 m (FIA 2022 Technical Regulations, Art. 3.4.1). The ratio is $\frac{2.000}{1.88} = 1.0638$. Therefore the estimated resulting area will be:

$$A_{2022} = A_{2026} \cdot 1.0638$$

$$A_{2022} = 0.744\ m^2$$

As the strength of the airflow doesn't change, nor the material or the inclination of the rear-wing, the pressure at the centre of the wing will be assumed to be as strong as in the 2026 model.

With this data, the Wake Rating for the 2022 regulated cars can be calculated:

$$Wake\ Rating = C_d \cdot A \cdot p_{max.}$$

$$Wake\ Rating = 0.744 \cdot 1 \cdot 4589.86$$

$$Wake\ Rating = 3414.86$$

In conclusion, the Wake Rating on the 2026 car, which takes into account its drag coefficient, cross-sectional area and the maximum pressure felt by the rear-wing, all of which were found via Computational Fluid Dynamics, is lower than the value calculated for the 2022-regulated cars. Which supports the thesis that the FIA has achieved its main goal, reducing the car's wake effects to promote closer racing. If we compare the numbers themselves, we can compute what the reduction of Wake Rating is:

$$Wake\ Effect\ \%\ Reduction = \frac{3414.9 - 2176.3}{3414.9} = \frac{1238.6}{3414.9} = 0.363 \approx 36\%$$

Therefore, the conclusion of this paper is that the 2026 regulations will bring a substantial reduction in the wake effects.

This conclusion aligns with the comparison of both physical wakes, visualized using the CFD simulations.

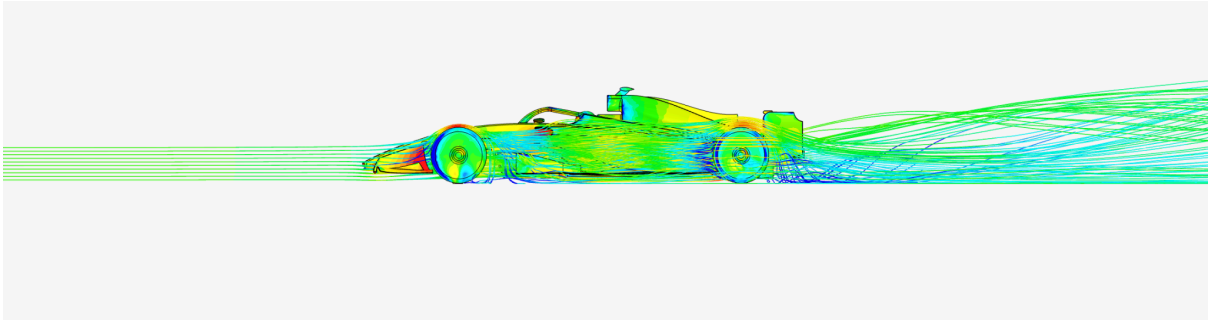


Figure 7) Visualization of the wake in the 2026 Model



Figure 8) Visualization of the wake in the 2023 Model

As it can be visually appreciated, the wake from Figure 8) goes immediately over the rear wing, while on Figure 7), the wake grows in a more gradual, controlled manner, suggesting that it is less energetic and supporting the conclusion made, which states that the wake effect created by the 2026 car is less energetic than the 2022-regulated cars.

Possible Extensions

To extend this investigation, simulations at different speeds could have been conducted, to determine how changes in velocity affect each of the model's characteristics. Moreover, the variation of aerodynamic components within a same model would allow this investigation to dig deeper into aerodynamic tweaks in both eras of modern motorsport.

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